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MINISTRY OF ELECTRONICS AND INFORMATION TECHNOLOGY

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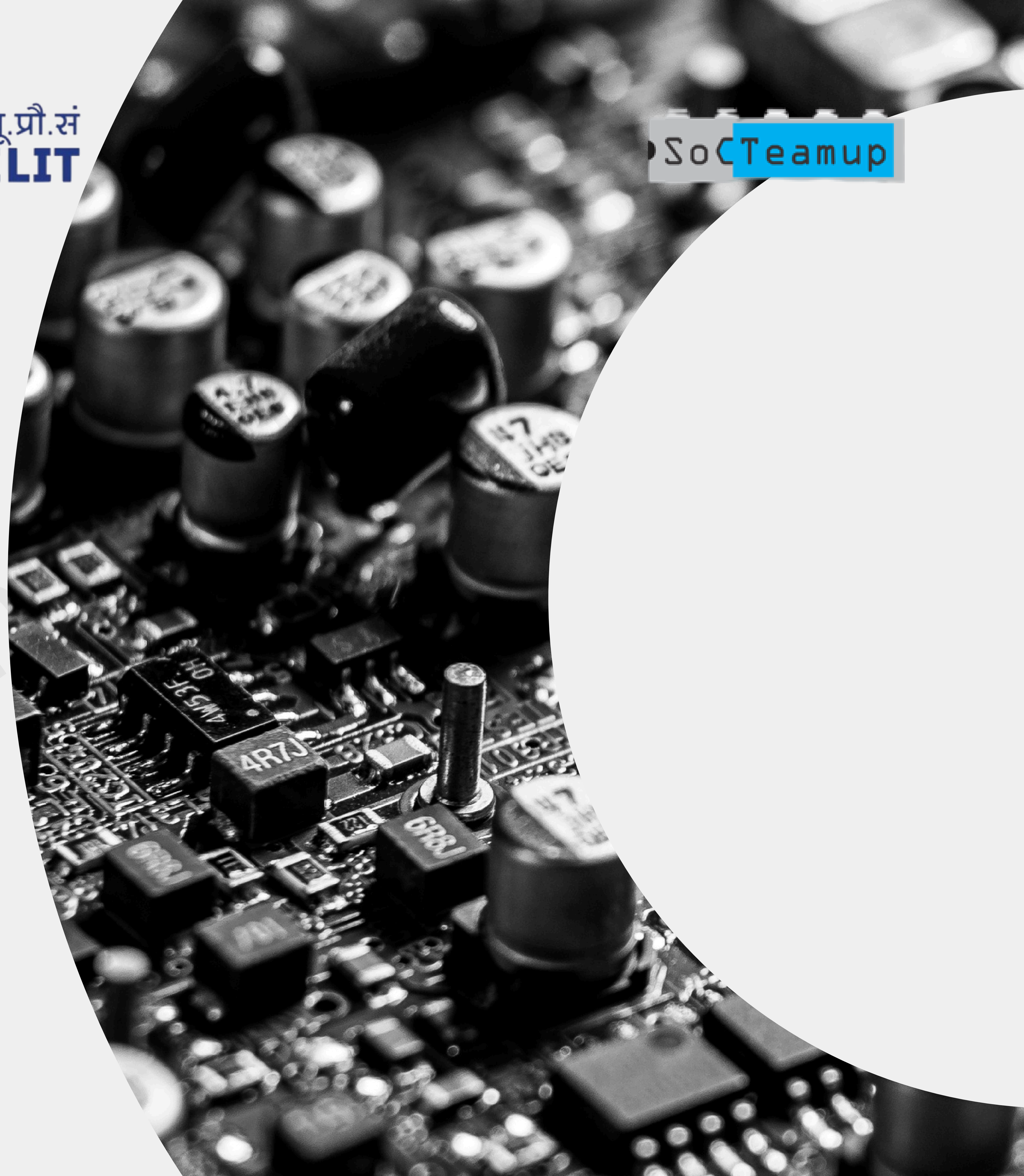
SoCTeamup

Introduction to Physical Signoff Formats



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Outline

1	What is SPEF?
2	Why Parasitics Need to Be Exchanged
3	When is SPEF Generated?
4	SPEF in the Signoff Flow
5	SPEF vs. SDF vs. DSPF
6	Structure of a SPEF File
7	Units and Netlist Format in SPEF
8	Example: SPEF Header and Units
9	Example: Parasitic Values for a Net
10	Lumped vs. Coupled Capacitance
11	RC Extraction Tools
12	How STA Uses SPEF
13	Common Issues and Debugging
14	Tools Supporting SPEF
15	Summary and Best Practices

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What is SPEF?

SPEF stands for Standard Parasitic Exchange Format.

It captures:

- Resistance (R) and Capacitance (C) values of interconnects
- For each net in the layout

It is used by STA tools to analyze real post-layout delays.

Why Parasitics Need to Be Exchanged

Wires introduce delay due to:

- Resistance (R) along the path
- Capacitance (C) between nets and to ground

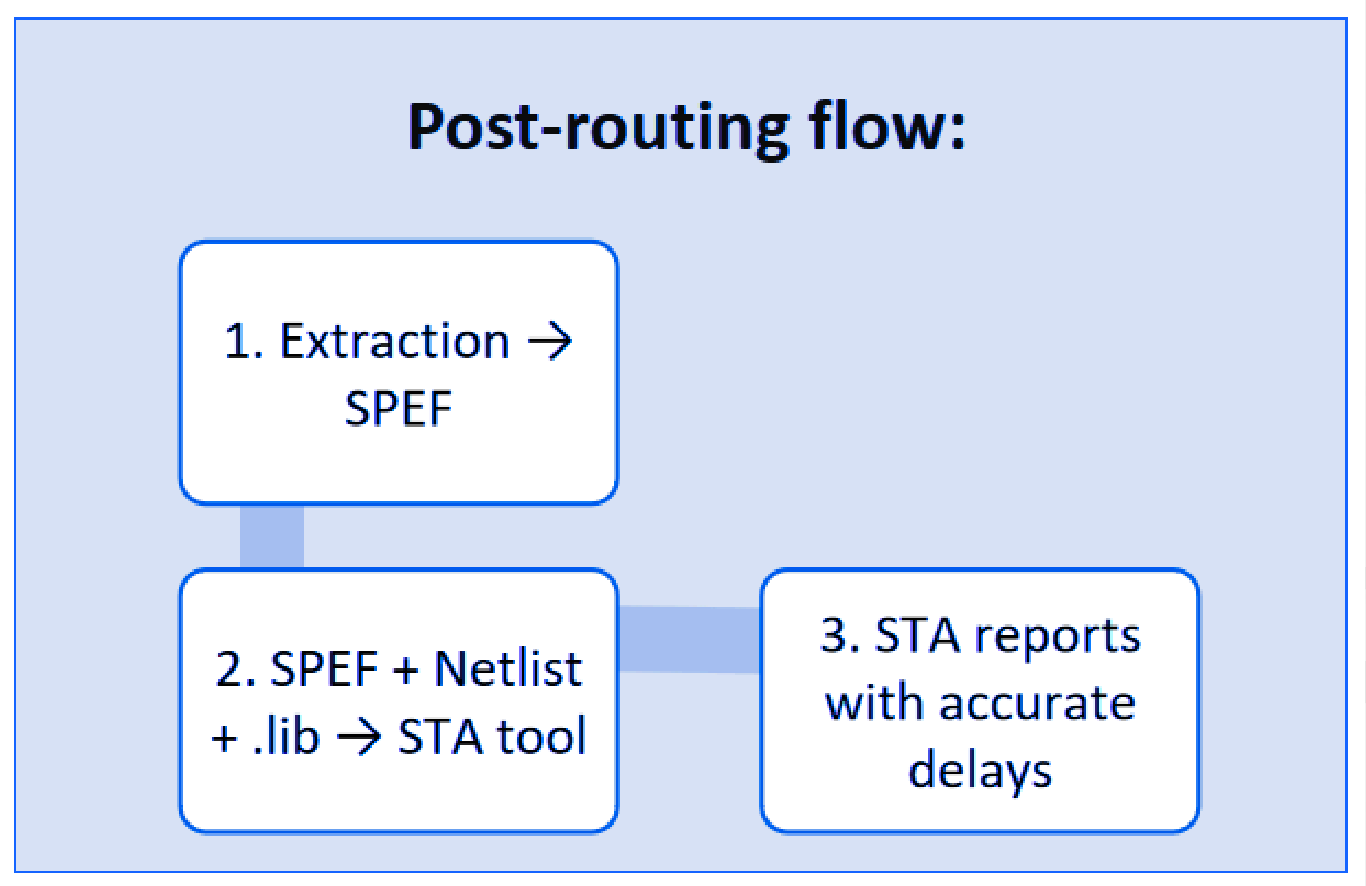
Tools extract parasitics → STA tools read them using SPEF → Timing analysis becomes accurate.

When is SPEF Generated?

SPEF is created after:
• Layout routing is complete • RC extraction is run

It reflects the actual physical routing and wire geometry in the chip.

SPEF in the Signoff Flow



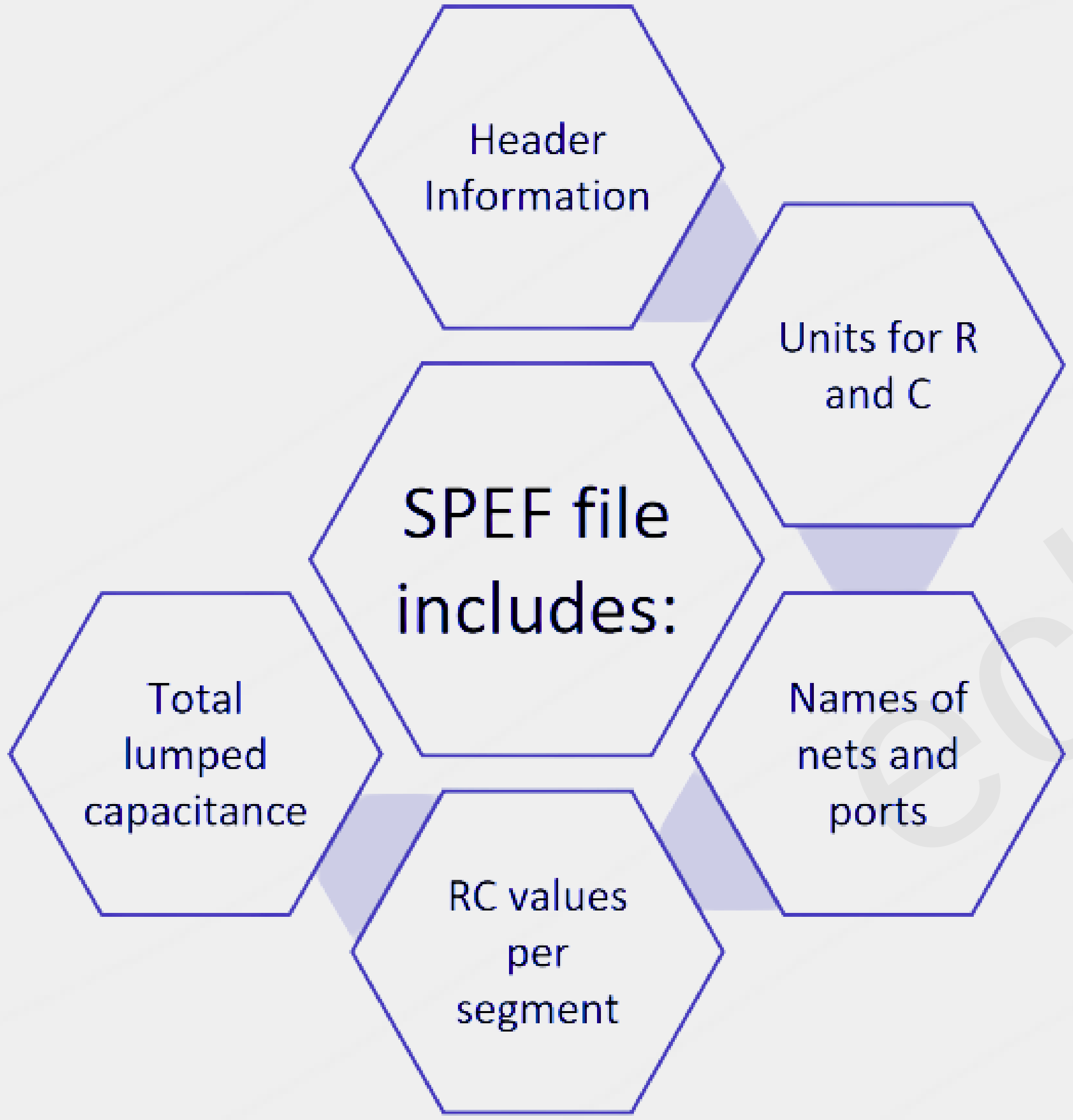
SPEF is critical for final timing and power signoff.

SPEF vs. SDF vs. DSPF

SPEF:	• RC parasitics (resistance, capacitance)
SDF:	• Timing delays for paths (setup/hold)
DSPF:	• Detailed parasitic format (transistor-level)

SPEF is most commonly used for gate-level STA.

Structure of a SPEF File



Units and Netlist Format in SPEF

SPEF uses:

- *C_UNIT: Farads (e.g., 1e-15 = 1 fF)
- *R_UNIT: Ohms (e.g., 1 = 1Ω)
- *D_NET: Net name and total capacitance

Hierarchical names are used like U1/A or CLK/Y

Example – SPEF Header and Units

Example:

```
*SPEF "IEEE 1481-1998"
*DESIGN "top_level"
*DATE "2024-06-30"
*VENDOR "OpenROAD"
*C_UNIT 1.0e-15
*R_UNIT 1.0
```

Example – Parasitics for a Net

Example:

```
*D_NET CLK 0.275
*CONN
*I CLK
*L U1/A
*CAP
1 U1/A 0.075
2 CLK 0.200
*RES
1 1 2 60.0
```

Total Cap: 0.275fF
Resistance between U1/A and CLK: 60Ω

Lumped vs. Coupled Capacitance

Lumped C:

- Total capacitance on a node

Coupled C:

- Capacitance between neighbouring nets

Coupling is important for noise and crosstalk analysis.
SPEF can include both types.

RC Extraction Tools

Common extraction tools:	Synopsys StarRC
	Cadence Quantus
	OpenRCX (OpenLane)

They analyze post-layout data and generate SPEF files.

How STA Uses SPEF

STA tools like PrimeTime or OpenSTA:

- Read netlist + SPEF
- Compute path delays using Elmore or π models
- Evaluate setup and hold checks

SPEF makes STA post-layout aware.

Common Issues and Debugging

Mismatch between netlist and SPEF hierarchy

Missing caps → inaccurate delay

Incorrect units → wrong timing

Always check:

- SPEF parser logs
- Units and net names

Summary and Best Practices

Summary:

- SPEF provides post-layout R and C info for nets
- Used by STA tools for accurate timing signoff

Best Practices:

- Always generate SPEF after detailed routing
- Match netlist and SPEF hierarchy
- Use accurate extraction settings

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Thank You